

CSE427 Section 5 Quiz 2

Fall 2025

Duration: 25 Minutes

Name:

ID:

Question [10 Points]

Suppose, you are given the data below and you want to classify the labels in Y using the Naïve Bayes Algorithm. Here, X_1 , X_2 , X_3 , X_4 are the features.

Instance	X_1	X_2	X_3	X_4	Y
1	P	E	12.5	8.0	A
2	Q	F	20.0	7.5	B
3	P	G	15.0	9.0	C
4	Q	E	18.5	7.0	C
5	P	F	10.0	8.5	A
6	Q	G	17.0	9.5	C
7	P	E	13.5	7.5	A
8	Q	F	19.0	8.0	B
9	P	G	14.0	9.0	A
10	Q	E	16.5	8.5	C

Suppose, you have a new instance: $x_{\text{new}} = (P, G, 11.5, 8.5)$

What will be the predicted class for x_{new} ?

If any of the probabilities are zero, add a small epsilon value ($\epsilon = 10^{-10}$) in both numerator and denominator.

Solution

$$P(A) = 4/10 = 0.4$$

$$P(X_1 = P \mid A) = 4/4 = 1$$

$$P(X_2 = G \mid A) = 1/4 = 0.25$$

$$P(X_3 = 11.5 \mid A) \approx 0.2097$$

$$P(X_4 = 8.5 \mid A) \approx 0.6457$$

$$P(B) = 2/10 = 0.2$$

$$P(X_1 = P \mid B) = (0+\epsilon)/(2+\epsilon) \approx 5 \times 10^{-11}$$

$$P(X_2 = G \mid B) = (0+\epsilon)/(2+\epsilon) \approx 5 \times 10^{-11}$$

$$P(X_3 = 11.5 \mid B) \approx 2.05 \times 10^{-56}$$

$$P(X_4 = 8.5 \mid B) \approx 0.0177$$

$$P(C) = 4/10 = 0.4$$

$$P(X_1 = P \mid C) = 1/4 = 0.25$$

$$P(X_2 = G \mid C) = 2/4 = 0.5$$

$$P(X_3 = 11.5 \mid C) \approx 4.72 \times 10^{-5}$$

$$P(X_4 = 8.5 \mid C) \approx 0.4265$$

$$P(A \mid X_{\text{new}}) = 0.4 * 1 * 0.25 * 0.2097 * 0.6457 \approx 0.01354$$

$$P(B \mid X_{\text{new}}) = 0.2 * (5 \times 10^{-11}) * (5 \times 10^{-11}) * (2.05 \times 10^{-56}) * 0.0177 \approx 1.82 \times 10^{-79}$$

$$P(C \mid X_{\text{new}}) = 0.4 * 0.25 * 0.5 * (4.72 \times 10^{-5}) * 0.4265 \approx 1.01 \times 10^{-6}$$

Normalized: (Optional)

$$P(A \mid X_{\text{new}}) \approx 0.9999$$

$$P(B \mid X_{\text{new}}) \approx 0$$

$$P(C \mid X_{\text{new}}) \approx 0$$

The predicted class will be A.